

Databases

Assignment 1 - Clear Away (CA)

Assignment 1 Checklist

Required Actions:

- ☐ Carefully checked the Marking Rubric in the assignment document so that you are aware of the mark allocation
- ☐ Task 1: Conceptual Model
 - ☐ Followed the unit conceptual model notation requirements as listed in your “Conceptual Modelling” Applied material
 - ☐ All entities and relationships on page 2-3 of the brief are included; *nothing extra has been added*
 - ☐ KEYs clearly indicated
 - ☐ All relationships have minimum and maximum cardinality shown, are indicated as identifying or non-identifying and have appropriate labels
 - ☐ Exported the PDF as an A4 single portrait page
- ☐ Task 2: Normalisation
 - ☐ Normalised the *Collection* report from UNF to 3NF, showing all stages and dependencies at each step (i.e. partial dependencies in 1NF, transitive dependencies in 2NF, full dependencies in 3NF). Included all candidate keys for each relation at 1NF
 - ☐ Included all attributes shown on the form; *no extra attributes added*
 - ☐ Did *not* add any surrogate keys in normalisation
 - ☐ Indicated PK's using underline of the PK attributes
 - ☐ Completed attribute synthesis, if required
 - ☐ Checked naming consistency between Conceptual Model and Normalisation
- ☐ Task 3: Logical Model
 - ☐ Followed the unit logical model notation/requirements:
 - ☐ Used Crow's foot/Information Engineering notation for the logical model
 - ☐ Does *not* show data types and sizes
 - ☐ Does show legend on the logical model, including your name/ID
 - ☐ Does show a label for each relationship
 - ☐ Added common prefix for all attributes in each relation (e.g. emp_no, emp_name)
 - ☐ Added comments for all attributes (used comments in RDBMS)
 - ☐ Added at least one surrogate key to the logical model and added an explanation of why it was chosen to the assumptions document
 - ☐ No relation with more than two attributes in the key remains (if more than two, a surrogate must be added)
 - ☐ Included all required check clauses and lookup tables
 - ☐ Integrated FULL normalisation results (final 3NF) into the logical model - all relations and attributes
 - ☐ Checked naming consistency between Conceptual Model, Normalisation and Logical Model
 - ☐ For each relation, all attributes are shown on the diagram (no downward-pointing diamond displayed to show some attributes are hidden)

- ❑ Made sure there are no data anomalies in the final logical model (i.e., all relations are in 3NF)
- ❑ Schema File
 - ❑ Has extension .sql
 - ❑ Included DROP table commands at the head of the schema file
 - ❑ Has not been edited other than to add a header (student details) and SPOOL/ECHO commands
 - ❑ Captured run of schema file via SPOOL/ECHO commands, and does not show any errors except for the drop table statements

Required files have been pushed to the FIT GitLab server (at least three pushes of Conceptual Model, three pushes of Normalisation and six pushes of Logical Model):

- ❑ ca_conceptual.pdf plus the source file
- ❑ ca_normalisation source file (source file in repo if MS Word/Pages) and final PDF of this document on GitLab
- ❑ ca_logical.pdf (check this is your final model and produced via File - Print Diagram - To PDF File from within Data Modeller, do not use screen capture)
- ❑ ca_model project folder, which includes:
 - ❑ the ca_logical.dmd file,
 - ❑ and the ca_logical folder

You must push this project folder at least six times while you are drawing the model with the Oracle Data Modeler

- ❑ ca_schema.sql
- ❑ ca_schema_output.txt
- ❑ ca_assumptions source file (source file in repo if MS Word/Pages) and final PDF of this document on GitLab

Finally, seven individual files *have been submitted to Moodle* (DO NOT zip these 7 files together). ONLY the files submitted to Moodle can/will be marked:

1. ca_conceptual.pdf
2. ca_normalisation.pdf
3. ca_logical.pdf
4. ca_model.zip, which includes:
 - the .dmd file,
 - and the model folder.
 - You **MUST ensure that the zip archive of the model is tested for completeness, i.e. includes both logical and relational models** (unzipped and opened in a new location - see video: Preparing Files for Submission in on Ed under the lesson "Oracle Data Modeler Support Videos")
5. ca_schema.sql
6. ca_schema_output.txt
7. ca_assumptions.pdf